

Diabetes 130-US Hospitals Data Cleaning Logs

Professional cleaning summary for the healthcare readmission dataset.

Field	Details
Dataset	Diabetes 130-US Hospitals for Years 1999-2008
Source files	diabetic_data.csv and IDS_mapping.csv extracted from diabetes.zip
Raw size	101,766 rows and 50 columns
Data types	13 integer columns and 37 object/categorical columns
Memory usage	Approximately 38.8 MB before cleaning
Target variable	readmitted with labels: NO, >30, <30
Cleaning tool	Python with Pandas and NumPy

Objective

The goal of this cleaning work was to prepare raw hospital encounter data for reliable analysis and future machine learning. The raw dataset contained categorical healthcare fields, placeholder missing values written as '?', age ranges, diagnosis codes, medication status columns, and a three-class readmission target.

Dataset Download and Loading

- Downloaded the dataset zip file using wget.
- Extracted diabetic_data.csv and IDS_mapping.csv from the zip file.
- Loaded diabetic_data.csv into a Pandas DataFrame.
- Checked the working directory to confirm that the CSV files were available.

Initial Data Understanding

Check	Observation	Reason
Shape	101,766 rows x 50 columns	Confirmed the dataset size before cleaning.
Column review	IDs, demographics, admission IDs, diagnosis columns, lab results, medications, and readmitted target	Identified columns that need encoding, missing-value handling, or validation.
Data types	int64: 13 columns; object: 37 columns	Most columns are categorical and need transformation before modelling.
Missing values	max_glu_serum: 96,420 nulls; A1Cresult: 84,748 nulls	These columns need deliberate handling instead of random row deletion.
Target distribution	NO: 54,864; >30: 35,545; <30: 11,357	The target has three readmission classes.

Cleaning Strategy

The raw data uses '?' as a missing-value placeholder in several categorical columns. The cleaning strategy was to convert those placeholders into proper missing or unknown values, encode important categorical fields, remove invalid categories, and keep a clear record of every transformation.

Area	Cleaning decision
ID columns	Keep encounter_id and patient_nbr; verify missing values and int64 datatype.
Demographics	Clean race, gender, and age; convert them into numeric form.
High-missing fields	Drop or document columns such as weight and payer_code when they mostly contain '?' values.
Diagnosis fields	Replace '?' with Unknown instead of deleting rows.
Medication fields	Map No, Steady, Up, and Down consistently across medication columns.
Target column	Convert readmitted labels into numeric classes.

Detailed Cleaning Log

Step	Column(s)	Cleaning performed and reason
1	encounter_id	Checked datatype and missing values. Kept/converted as int64 because it is the encounter-level identifier.
2	patient_nbr	Checked missing values and kept/converted as int64 because it identifies patients across encounters.
3	race	Checked unique values, replaced '?' with NaN, filled missing race using mode, then encoded categories into numeric values.
4	gender	Removed Unknown/Invalid records and mapped Male=1, Female=0 to make the column usable for analysis.
5	age	Converted age buckets such as [30-40) into midpoint values such as 35 so the feature becomes numeric.
6	weight	Marked for removal because most values are '?' placeholders, making reliable imputation difficult.
7	admission_type_id, discharge_disposition_id, admission_source_id	Checked datatype and kept as numeric ID fields because they were already encoded.
8	diag_1, diag_2, diag_3	Replaced '?' with NaN and filled missing diagnosis values with Unknown to preserve rows without inventing codes.
9	change	Should map No=0 and Ch=1. Important note: this column uses Ch, not Yes.
10	diabetesMed	Mapped No=0 and Yes=1 to convert a binary feature into numeric form.
11	readmitted	Mapped NO=0, >30=1, and <30=2 to create numeric target labels.

Categorical Encoding Reference

Column	Original values	Cleaned / encoded values
race	Caucasian, AfricanAmerican, Asian, Hispanic, Other, ?	Caucasian=1, AfricanAmerican=2, Asian=3, Hispanic=4, Other=5; '?' filled with mode before mapping.
gender	Female, Male, Unknown/Invalid	Female=0, Male=1; Unknown/Invalid removed.
age	[0-10), [10-20), ..., [90-100)	5, 15, 25, 35, 45, 55, 65, 75, 85, 95.
change	No, Ch	No=0, Ch=1. Using Yes here would leave Ch unmapped.
diabetesMed	No, Yes	No=0, Yes=1.
readmitted	NO, >30, <30	NO=0, >30=1, <30=2.

Important Data Quality Notes

- Pandas isnull() does not catch '?' placeholders. Replace '?' first, then check missing values again.
- max_glu_serum and A1Cresult contain many missing values. Treat them carefully as categorical lab-result fields.
- The change column should be mapped with Ch, not Yes. This is a common mistake in this dataset.
- Diagnosis columns can be left as Unknown or grouped into broader diagnosis categories for modelling.

Recommended Additional Checks

Check	Suggested validation
Duplicates	Run df.duplicated().sum() and check duplicate encounter_id values.
Numeric ranges	Validate time_in_hospital, num_lab_procedures, num_procedures, num_medications, and diagnosis counts.
High-cardinality columns	Review medical_specialty and payer_code before dropping, grouping, or encoding them.
Medication columns	Map No, Steady, Up, and Down consistently across all medication-related columns.
Final target	Confirm readmitted contains only 0, 1, and 2 after mapping.

Final Validation Checklist

Validation item	Expected result
Final shape	Print final rows and columns after all cleaning steps.
Missing values	No unexpected missing values in encoded columns.
Datatypes	ID columns and encoded features should be numeric where needed.
Gender	No Unknown/Invalid values should remain.
Race	No '?' values should remain.
Age	Age should contain numeric midpoint values.
Target	readmitted should contain labels 0, 1, and 2 only.
Saved output	Cleaned dataset should be saved as CSV for analysis or model building.

Suggested Final Notebook Cell

```
# Final validation
print(df.shape)
print(df.isnull().sum())
print(df.dtypes)
print(df.duplicated().sum())
print(df['readmitted'].value_counts())

# Save cleaned dataset
df.to_csv('diabetes_cleaned_data.csv', index=False)
```

Conclusion

The Diabetes 130-US Hospitals dataset was cleaned by identifying placeholder missing values, transforming demographic and target fields, handling diagnosis columns, and preparing categorical columns for analysis. These cleaning logs document what was checked, what was changed, and why each step was required. The cleaned dataset is now more suitable for exploratory analysis, dashboard work, and machine learning experiments related to hospital readmission.